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Towards Prescriptive Learning Analytics by using Counterfactual XAI Explanations from Educational Time Series Dataset

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Abstract

Early-warning models based on student interaction data are widely used in learning analytics to identify learners at risk in online education. Despite their predictive accuracy, the adoption of such models in practice remains limited, partly due to the lack of explanations that support instructors in understanding model behavior and reasoning about potential interventions over time.

This work presents a framework for saliency-guided counterfactual explanations of multivariate educational time series. The framework integrates temporal attribution methods with two novel counterfactual generation strategies that explicitly restrict behavioral modifications to the most influential time steps and variables identified by predictive models. By embedding attribution directly into the counterfactual generation process, the proposed methods aim to produce sparse, temporally coherent, and interpretable behavioral alternatives.

The framework is evaluated on the Open University Learning Analytics Dataset (OULAD) using early-course prediction models and is compared against state-of-the-art time-series counterfactual baselines. The results indicate that saliency-guided counterfactuals achieve improved sparsity and

validity while maintaining temporal plausibility, yielding behaviorally localized modifications that are perceived as easier to interpret.

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Keywords: Counterfactual Explanation, Explainable Artificial Intelligence, Educational Data Mining, Predicting Student Performance

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1. Introduction

The expansion of online learning has facilitated the collection of detailed data on students' sequential interactions, often in the form of clickstream logs that record student activity over time. These multivariate time series capture behavioural patterns that can be leveraged to predict educational risks, including dropout and academic failure [1, 2, 3]. Early-warning models based on such data have shown strong predictive performance, particularly when using deep learning architectures capable of modelling temporal dependencies. However, despite their accuracy, the adoption of these models in real educational contexts remains limited. One important reason is their inherent opacity, which constrains their practical applicability in settings where timely, interpretable, and actionable insights are essential to support instructional decisions [4, 5].

Explainable Artificial Intelligence (XAI) has emerged as a response to this challenge, aiming to make model behaviour more transparent and understandable to human stakeholders [6, 7]. In educational settings, attribution-based techniques such as SHAP or Integrated Gradients have been used to highlight which features or input segments contribute most to a given prediction [8, 9]. While these approaches can provide valuable insights into model sensitivity, they primarily offer retrospective explanations that answer why a prediction was made. They rarely address a more practical and pedagogically relevant question frequently posed by instructors: what could have been done differently, and at which moments in time, to achieve a better outcome?

Counterfactual explanations respond directly to this question by proposing minimal and realistic modifications to an input that would lead a predictive model to produce a different outcome [10]. In educational contexts, this form of explanation is particularly appealing, as it aligns with instructors' habitual reasoning about alternative learning trajectories and potential intervention points. A counterfactual scenario describes a hypothetical situation in which one or more behavioural variables are altered to examine their effect on predicted outcomes. For example, a student who fails to submit a key assignment early in a course may be predicted to fail, whereas a counterfactual trajectory in which the same student submits that assignment on time may lead the model to predict successful completion.

Despite this potential, most research in XAI has focused on tabular and image data, while time series—such as those generated in online learning environments—have received comparatively less attention [11, 12]. Unlike tabular data, which consist of independent rows and columns, educational time series

represent sequences of behaviours that unfold over time and are shaped by temporal dependencies and cumulative effects. In online education, student engagement is inherently dynamic, and identifying when particular actions matter can be as important as identifying which actions matter. XAI techniques applied to time series can help detect influential moments in a learning trajectory, opening new possibilities for understanding and supporting student engagement over time. However, even when temporal explanations are available, their complexity often limits their interpretability for educators, underscoring the need for systematic, user-oriented approaches that balance technical validity with pedagogical relevance.

Counterfactual explanations are especially promising in this regard, as small behavioural differences at specific moments in time can lead to substantial changes in model predictions. Nevertheless, counterfactual methods for time series remain underexplored, particularly in educational contexts. Existing approaches such as CoMTE [14] and TSEvo [15] have been developed for general-purpose temporal data and do not explicitly account for key characteristics of educational clickstreams, including course structure, feasibility constraints, and the need for interpretability in instructional decision-making. As a result, these methods often generate counterfactuals that are technically valid but difficult for educators to interpret, relate to pedagogical practice, or use as a basis for reflection.

In this paper, we propose two saliency-guided counterfactual generation methods designed specifically for multivariate educational time series. The core idea is to integrate temporal attribution directly into the counterfactual generation process, using saliency information to restrict behavioural mod-

ifications to the most influential time steps and variables identified by the predictive model. By making small, focused adjustments around these key moments, the resulting counterfactuals are sparser, more temporally coherent, and easier to interpret, while remaining valid with respect to the model's decision boundary. Rather than claiming direct educational impact or prescribing specific interventions, the aim of this work is to advance explainable learning analytics [16] by providing time-based, action-oriented explanations that support instructor reasoning, reflection, and sensemaking

The present study advances the field through four main contributions:

1. It conceptualizes the generation of counterfactual explanations for multivariate educational time-series data as a problem of supporting instructor reasoning about student engagement over time, explicitly addressing challenges related to temporal dependencies, feasibility constraints, and interpretability in educational contexts.
2. It introduces two saliency-guided counterfactual generation techniques that integrate temporal attribution into the counterfactual search process, enabling behavioural modifications to be focused on the most influential time steps and variables identified by predictive models.
3. It proposes a comprehensive quantitative evaluation framework for time-series counterfactual explanations, assessing multiple quality dimensions including validity, similarity, sparsity, plausibility, and computational cost, and uses this framework to compare the proposed methods with existing state-of-the-art approaches.
4. It reports exploratory qualitative feedback from university instructors, offering initial insights into the perceived interpretability and poten-

tial usefulness of temporally grounded counterfactual explanations as decision-support artifacts for reflective use in online teaching contexts.

The remainder of the paper is structured as follows. Section II reviews related work on explainability and counterfactual explanations in temporal and educational contexts. Section III formalizes the problem setting and defines actionability constraints for educational time series. Section IV presents the proposed saliency-guided counterfactual generation methods. Section V reports the experimental setup and quantitative results. Section VI discusses implications, limitations, and directions for future research, and Section VII concludes the paper.

2. Related Works

This section reviews prior research from three interrelated perspectives. The first concerns XAI methodologies and their advancements in addressing time-series datasets. The second perspective covers the application of explainability methods within the domains of EDM and LA. The last one focuses on counterfactual explanations, exploring their applications, potential, and challenges in education.

2.1. Explainable Artificial Intelligence for Time Series

The increasing complexity of machine learning models, especially deep neural networks, has raised concerns about their lack of transparency and explainability. Popular XAI methods, such as LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) provide local explanations by approximating the importance of specific input variables for a model's prediction. These techniques demonstrated high

effectiveness across tabular datasets and images, providing a natural understanding of a highly intricate, non-transparent decision-making process.

LIME, introduced by Ribeiro et al. [17], provides locally interpretable explanations. It fits a simple model around a complex one, making it easier to see how individual features contribute to a specific prediction. SHAP, developed by Lundberg and Lee [8], is based on Shapley values from game theory. Provides a clear, consistent measure of each feature's importance for model predictions at both local and global levels.

Nonetheless, these XAI techniques are relatively scarce for time-series data. Time-series data have additional complexities due to temporal dependencies and ordering restrictions, as well as underlying processes that change over time. [12, 11]. Therefore, recent work has focused on adapting attribution and saliency-based methods to temporal models to pinpoint critical time steps or segments of a series.

Several saliency maps and attribution methods[13], also known as time series relevance methods, have been proposed to demonstrate how different segments or points in a time series impact an AI model's predictions. These methods enable researchers and practitioners to gain a deeper understanding of the subject matter. One of the most basic attribution methods is Vanilla Gradients (GRAD) [9], which computes the gradient of the model output with respect to its input features. In time series, this indicates how small changes in specific time steps affect the prediction, but it can be noisy and sensitive to model saturation and its variations. To address this, Smooth Gradients (SG) [18] averages gradients over multiple noisy copies of the input, reducing visual noise and enhancing interpretability. SG is particularly

useful when sharp GRAD attribution maps obscure the temporal dynamics of the signal. Integrated Gradients (IG) [19] further improves robustness by accumulating gradients along a path from a baseline input (e.g., a zero or neutral sequence) to the actual input. This integration captures how the model's output evolves, helping to mitigate issues of saturation and improve attribution faithfulness. Additionally, Schlegel et al. [20] proposed a method to evaluate time series relevance methods using perturbation and sequence analysis. They examined the impact of different key segments of the series, focusing on how performance changes when these moments are altered.

2.2. Explainability in Learning Analytics and Online Education

In recent years, predicting academic performance and assessing dropout risk have gained increasing attention in educational research [1]. Various algorithms, ranging from logistic regression to decision trees, have been employed to identify risk factors that contribute to student outcomes [21]. Models like these have been widely used because they are simple and provide clear, understandable insights into which variables affect students' performance. However, with each passing year, the volume and complexity of education data grow. Consequently, more complex deep learning-based methods have been applied, including Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), and Long Short-Term Memory Networks (LSTMs) [22], which have become increasingly popular. These deep learning models excel at managing large and complex datasets, achieving superior accuracy in predicting academic performance and dropout risk [23, 24], particularly with time series [2, 3].

The main challenge is the "black box" nature of deep learning models;

hence, they are not applicable on a large scale in education because interpretability and transparency are required to build trust and facilitate decision-making by educators.

The increasing emergence of XAI in Educational Data Mining (EDM) may therefore be a promising avenue for future research [25]. Current EDM research has focused on developing increasingly complex models with the primary goal of enhancing model accuracy. However, when model predictions influence significant decisions in an educational setting, transparency becomes a paramount condition for the model's acceptance. Therefore, in the field of educational data mining, XAI methodologies have found growing applicability for deriving explanations of student behavior related to success or dropout, and for assisting course instructors in understanding model predictions.

Although very few studies have specifically addressed the explainability of prediction models in education [4], recent research has started to explore how XAI techniques can provide insight into the factors that drive student performance and retention in online learning environments. Kizilcec et al. [26] pointed out that XAI is capable of revealing predictive behavioural patterns of online course dropout rates through intuitive explanations that are fundamental for devising effective and timely interventions. Other studies have emphasized how XAI can help identify key characteristics that affect student performance, empowering educators to implement more targeted strategies [5].

However, there are significant shortcomings in the use of general-purpose XAI techniques in education. Many of them did not adequately address

the temporal dependencies and contextual constraints typical in the learning situation. Although recent efforts have incorporated attention mechanisms and used information graphs to improve interpretability [27], several limitations persist. Specifically, adapting general-purpose XAI methods, such as LIME, to educational data has revealed substantial shortcomings. For example, LIME demonstrates considerably lower explanatory power in educational environments than in other domains, often failing to account for the complex temporal dependencies and nuanced behavioural patterns that characterize student learning trajectories. Current XAI methodologies lack the necessary granularity and contextual sensitivity required for educational applications. Consequently, there is a pressing need to develop advanced context-aware XAI techniques that directly address the intricate and dynamic nature of educational time series data [28].

2.3. Educational Counterfactual Explanations

Counterfactuals give instructors and institutions a chance not only to unravel a model prediction, but also to ask the question "What would have had to happen for the result to have been different?" This reasoning opens the way for more pragmatic and actionable insights in the name of a better sense of fairness, trustworthiness, and cooperation between institutions, instructors, and students [29, 10, 30].

Counterfactual explanations have the benefit of efficiently mapping questions that instructors already consider. What would the student have done if they had started the course a week earlier? Would greater access to resources or more consistent participation have changed the final outcome? These are not abstract questions; they are the kind of questions that underlie real de-

cisions about whether and how they should intervene [31].

Counterfactual explanations have been extensively explored in areas such as healthcare and finance, but their use in education remains limited. The educational context poses challenges in terms of feasibility, interpretability, and temporal coherence. Most counterfactual methods leverage static student records, though learning and disengagement typically unfold over time as consistent behavioral patterns [25, 16, 32]. The question then becomes which educational settings are best suited for counterfactual analysis, which techniques are generalizable enough for use in a wide array of learning settings, and how to report outcomes in a way that is both useful for educators and easy for them to apply.

One disadvantage is the technical complexity of understanding such models. Algorithmic reasoning is not common knowledge among many instructors. In the improbable event that the results of a model are given in abstract or too technical a form, they may sit idle. Explanations must first be presented in straightforward terms and in a language that flows naturally in day-to-day teaching [33].

Recently, state-of-the-art counterfactual approaches for multivariate time series have been introduced. For instance, COMTE (Counterfactual Optimization for Temporal Explanations) [14] utilizes effective optimization methods to understand how various factors influence the results over time. Another related approach, TSEvo [15], combines time-series transformation with evolutionary algorithms to generate counterfactuals while maintaining proximity and sparsity constraints. mainly optimization-based and evolutionary methods. Despite their breakthroughs, they are too generic and do

not exploit specific properties of educational clickstreams, including both course design (e.g., hard constraints) and instructor interpretability; therefore, the solutions these methods provide, albeit feasible, may not be actionable from a pedagogical viewpoint.

However, the inclusion of temporal attribution in counterfactual generation is a largely unexplored avenue. Saliency maps can be leveraged post hoc to assess models, though they are rarely ordered to indicate where counterfactual changes should occur or even how they should manifest. This separation limits one of the most beneficial aspects of counterfactual explanations: their ability to create succinct, temporally relevant, and interpretable alternatives. [34, 35, 36, 37].

The reviewed literature highlights a gap between the descriptive, explainability methods, and explanations that can lead to actions in learning analytics. While attribution methods can provide useful information about the sensitivity of the model and counterfactual explanations seem to be a step in a promising direction for actionable alternatives, the current methods do not sufficiently capture the temporal and contextual nature of the data present in education.

The current study fills this void by proposing saliency-guided counterfactual generation approaches suitable for multivariate educational time series. By using temporal attribution to direct counterfactual behavior changes toward the most salient time steps and variables, the proposed methods aim to deliver temporally consistent, feasible, and interpretable counterfactual explanations in educational settings.

3. Problem Definition

This section formalizes the problem of generating counterfactual explanations for educational time series and defines the notion of actionability adopted in this work. The goal is to set explicit conceptual and practical constraints that guide the subsequent methodological design and ensure that generated counterfactuals are useful within educational contexts.

Let $X \in \mathbb{R}^{T \times D}$ denote a multivariate educational time series representing a student's interaction trajectory, where T is the number of time steps and D is the number of behavioral variables (e.g., resource access, assessment activity, or platform navigation events). A predictive model $f(X)$ outputs a target value (outcome of interest), such as an early-course risk or performance prediction.

For a given example X, a counterfactual explanation is an alternative sequence X' that leads the model to predict a different outcome, i.e., $f(X') \neq f(X)$. The counterfactual X' should differ from X as much as necessary, subject to feasibility constraints. In educational settings, this implies that counterfactuals should not only change the prediction, but also suggest realistic behavioral modifications that can inform intervention planning.

Unlike static data, educational time series exhibit temporal dependencies and cumulative patterns based on past actions. Therefore, counterfactual explanations should preserve the temporal order and plausible progression of learning behaviors.

3.1. Actionability in Educational Time Series

Actionability refers to how closely the suggested counterfactual change is consistent with what is realistically achievable in an authentic educational setting. In this work, actionability is treated as a primary design principle rather than a secondary evaluation metric.

Data modifications must obey the temporal constraints. That is, one cannot randomly displace behavioral events in time, and any manipulation should also be temporally oriented in relation to the original trajectory. In that case, the shifts depend on both course design and platform affordances. The classic example is that an increase in assessment-relevant behavior is legitimate only at a time step adapted to an assessment, whereas an abrupt increase or disembodied rise in behavior might be pedagogically meaningless.

Finally, counterfactuals should be small to avoid large, transitive behavior deviations. If the deviation is large-scale or large enough to affect many time steps, the instructor may misinterpret it quickly and find it irrelevant. Therefore, counterfactuals should be temporally localized and scale-limited to create small, yet impactful, deviations.

3.2. Temporal and Structural Requirements

Counterfactual explanations for educational time series also need to respect the timing and structure of online courses, along with which variables can actually be changed.

Changes to the data should respect temporal ordering. In other words, behavioral events cannot be moved around in time indiscriminately, and any transformation should be temporal, relative to the original trajectory. Then,

the changes are subject to course design and platform affordances. An obvious example is that a rise in behavior relevant to assessment is valid only at the time step when an assessment is present, whereas a sudden spike or isolated rise in behavior may lack pedagogical meaning.

In addition, counterfactuals should be limited in scale to prevent extreme deviations of transitive behavior. A wide-scale or large deviation that affects many time steps can be easily misinterpreted and become irrelevant to the instructor. Thus, counterfactuals should be temporally localized and limited in scale, focusing on small but effective changes.

3.3. Plausibility and Explainability Requirements

Beyond validity with respect to the predictive model, counterfactual explanations must also meet plausibility and interpretability criteria. Plausibility is defined as how similar the counterfactual trajectory is to observed behavior patterns among actual students, whereas interpretability is defined as how easily educators can comprehend and rationalize the suggested changes.

In educational scenarios, the concept of plausibility intersects with temporal coherence and accordance with standard patterns of student behavior. Certain counterfactuals may undermine confidence in the model if they suggest disconnected or inconsistent behaviors, even though they can still influence its predictions. The concept of interpretability favors counterfactual explanations that identify a limited number of significant changes in behavior during distinct time intervals.

These claims imply the need for counterfactual explanations that focus on the when and where of changes, rather than on increases or decreases in a particular quantity across the entire time series.

3.4. Desired Properties of Educational Counterfactual Explanations

Based on the above considerations, counterfactual explanations for educational time series are expected to satisfy the following properties:

- Validity: the modified sequence leads to a different model prediction.
- Sparsity: only a limited number of variables and time steps are modified.
- Temporal coherence: changes respect the data's sequential structure.
- Plausibility: proposed behaviors resemble realistic student trajectories.
- Interpretability: explanations are readily understandable to educators and can be linked to potential instructional actions.

These properties refer to the constraints and objectives that guide the counterfactual generation methods introduced in the following section.

4. Methodology

This section presents a saliency-guided framework for generating counterfactual explanations from multivariate educational time series. As shown in Figure 1, the framework integrates predictive modeling, temporal attribution, and counterfactual reasoning to produce sparse, temporally coherent, and interpretable behavioral alternatives. Two complementary counterfactual generation strategies are introduced: a saliency-guided local search method and a saliency-guided evolutionary approach.

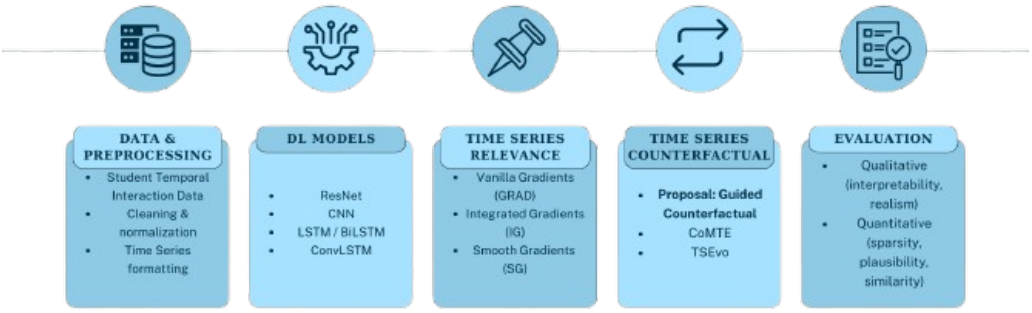


Figure 1: Methodology.

4.1. Framework Overview

A counterfactual explanation provides an alternative sequence X' for a particular example X such that the model produces a different prediction for X'. In educational settings, counterfactuals should not only change the predicted outcome but also propose feasible behavioral adjustments that can support intervention planning. Since educational data are sequential and evolving, counterfactual explanations should respect the temporal and cumulative nature of learning behaviors.

4.2. Predictive Model and Notation

Let $f : \mathbb{R}^{T\times D} \to \mathcal{Y}$ denote a predictive model trained on educational time series data, where T is the number of time steps and D is the number of behavioral variables. Given an input sequence $X \in \mathbb{R}^{T\times D}$, the model outputs a prediction score or a class label describing the expected student outcome.

While gradient-based attribution is required to estimate temporal saliency, the proposed framework remains independent of the specific architecture or training procedure of the predictive model once saliency information is avail-

able. Counterfactual generation is performed post hoc, reflecting the behavior of the deployed predictor.

4.3. Temporal Attribution via Saliency Maps

To identify influential temporal regions, gradient-based attribution methods are applied to the trained predictive model. For an input sequence X, an attribution method returns a saliency matrix

$$S \in \mathbb{R}^{T \times D},$$

where each element $S_{t,d}$ quantifies the sensitivity of the model output to changes in variable d at time step t.

Integrated Gradients (IG) was selected as the primary attribution technique due to its robustness to gradient saturation and its ability to capture cumulative effects along a baseline-to-input path. Vanilla Gradients (GRAD) and SmoothGrad (SG) are also considered for comparative analysis. Saliency values are normalized to enable ranking of time–variable pairs by relevance.

For counterfactual generation, only time-variable pairs whose saliency exceeds a predefined percentile threshold. This yields a binary relevance mask $M \in \{0,1\}^{T \times D}$ that determines which regions of the sequence are eligible for modification. Percentile-based thresholds are used to balance interpretability and sparsity by progressively restricting changes to the most salient temporal regions, ensuring that counterfactuals remain focused and cognitively manageable for instructors.

4.4. Saliency-Guided Local Search Counterfactuals

The saliency-guided local search strategy generates counterfactual explanations by restricting candidate modifications to salient time-variable pairs.

Given an input sequence X and its saliency matrix S, only entries with $M_{t,d} = 1$ (i.e., saliency above the percentile threshold) are perturbed.

Candidate counterfactuals are generated via bounded random perturbations applied to the selected regions and retained only if they induce a change in the predicted class. Among all valid candidates, LCSF select the counterfactual that minimizes both the distance to the original sequence and the number of modified elements. This procedure enables efficient exploration of the counterfactual space while preserving proximity to the original learning trajectory.

As shown in Algorithm 1, the Guided Local Search mechanism selectively alters key points with a user-defined tolerance *tol*. Only instances that successfully switch the predicted class are retained and the optimal one, based on distance and sparsity, is returned.

Algorithm 1 Guided Local Search Counterfactual

Require: Predictive Model M, Original Instance X, Target Class C_t, tolerance tol, number of samples n, percentile p

$R \leftarrow$ Relevances in X
$threshold \leftarrow$ Calculate relevance threshold based on p
$CFs \leftarrow \emptyset$ ▷ Initialize empty list
for $i \in 1 \rightarrow n$ **do**
    $X_{random} \leftarrow X$
    for each moment t and feature v **do**
        if $R[t, v] > threshold$ **then**
            $X_{random}[t, v] \leftarrow X[t, v] + random(0, tol)$
        end if
    end for
    $Pred \leftarrow M.predict(X_{random})$
    if $Pred = C_t$ **then**
        $CFs \leftarrow CFs \cup X_{random}$
    end if
end for
$CF_{best} \leftarrow$ Select best CF
return CF_{best}

4.5. Saliency-Guided Evolutionary Counterfactuals

The second approach extends an evolutionary counterfactual method by integrating saliency constraints directly into the search process. Candidate sequences are initialized by perturbing actionable variables at salient time steps and iteratively refined using evolutionary operators such as mutation

and selection.

Saliency guidance is incorporated in two ways: (i) mutation operations are biased toward highly relevant temporal regions, and (ii) fitness evaluation penalizes modifications outside salient segments. The resulting fitness balances prediction confidence, similarity to the original sequence, sparsity, and plausibility, while additionally enforcing saliency-based constraints.

Although evolutionary strategies typically require more computation than local search, they can explore a broader solution space and produce diverse counterfactual alternatives. Both proposed methods share the central principle of integrating temporal attribution directly into counterfactual generation rather than treating explanations as a purely post hoc step.

5. Experimental Setup

This section describes the experimental setup used to evaluate the proposed saliency-guided counterfactual generation methods, including the dataset, predictive models, attribution techniques, baseline methods, and evaluation metrics.

5.1. Data and Processing

Experiments were conducted using the Open University Learning Analytics Dataset (OULAD) [38], a widely used benchmark in EDM that provides longitudinal records of student interactions within a virtual learning environment (VLE). These interactions include participation in course content, forums, and assessments.

First, only clickstream features were selected, and then, to identify the most informative variables, a permutation-based, model-agnostic feature-

importance method was used [39]. This technique quantifies the influence of individual features by evaluating the change in model performance after randomly permuting each variable.

Figure 2 shows the results of this evaluation, and Table 1 lists the final variables selected, representing a diverse range of interaction types. For each student, a multivariate time series is constructed covering the first 90 days of course activity. This early observation window aligns with prior work on early risk detection [40].

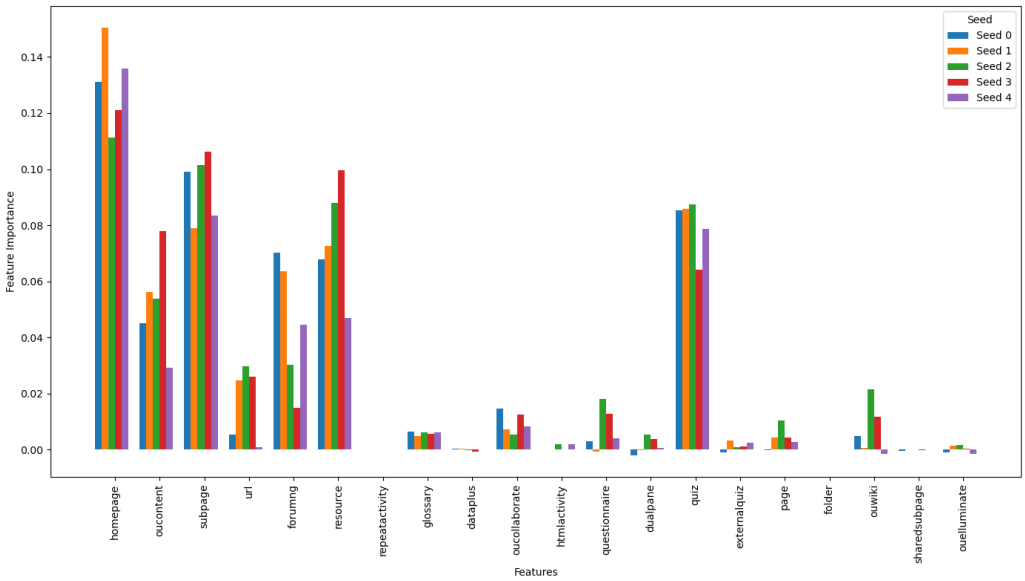


Figure 2: Feature Importance.

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Table 1: Selected Student Features

Feature	Description
HomePage	Engagement with the course homepage.
Oucontent	Access to assignment content.
SubPage	Clicks on internal course pages.
URL	Visits to linked external media.
Forum	Participation in discussion forums.
Resource	Access to learning materials (e.g., PDFs).
Quiz	Interactions with course assessments.

The dataset includes outcome labels grouped into four categories: *Withdrawn*, *Fail*, *Pass*, and *Distinction*. Following conventions in early intervention research [1, 21], these labels were aggregated into two broader classes: *At-Risk* (Withdrawn + Fail) and *Good Performance* (Pass + Distinction), resulting in a class distribution conducive to balanced learning.

5.2. Deep Learning Predictive Models

Several deep learning architectures suitable for sequential data are evaluated, including LSTM, Bidirectional LSTM (BiLSTM), CNN, and ResNet [41]. Model training is performed using five-fold cross-validation with an 80/20 train–test split. The Adam optimizer and early stopping are used to mitigate overfitting.

Model performance is assessed using Accuracy, F1-score, and Area Under the ROC Curve (AUC). The best-performing model is selected for subsequent

attribution and counterfactual analyses. The counterfactual framework operates on trained models without modifying their internal parameters, assuming access to gradient-based temporal attribution signals.

To ensure a fair comparison across predictive models, a unified training configuration was applied in all experiments, as detailed in Table 2.

Table 2: Training configuration for predictive model comparison.

Item	Value
Observation window	First 90 days (T=90)
Task	Binary classification (At-Risk vs Good Performance)
Split protocol	5-fold StratifiedKFold CV, shuffle=True, random_state=42
Optimizer	Adam
Loss	Binary cross-entropy
Batch size	8
Max epochs	100
Early stopping	patience=10, monitor=val_loss, restore_best_weights=True
Class imbalance	Balanced class weights (scikit-learn compute_class_weight)
Decision threshold	0.5
Padding	Post-padding to length 90 (value = -10.0)

Table 3 summarizes the main architectural components of each predictive model considered in the comparison.

5.3. Temporal Attribution via Gradient-Based Relevance

To understand how the models generate predictions across temporal sequences, three saliency-based attribution methods tailored to the time series

Table 3: Predictive model architectures used in the experiments.

Model	Architecture	Output
LSTM	LSTM(128, return_seq=True) → LSTM(64)	Dense(1, sigmoid)
BiLSTM	BiLSTM(128, return_seq=True) → BiLSTM(64)	Dense(1, sigmoid)
CNN	Conv1D(128, k=3) → MaxPool → Conv1D(256, k=3) → MaxPool → Flatten	Dense(1, sigmoid)
ResNet	Conv1D(64, k=7, s=2) + BN + ReLU + MaxPool → Residual blocks (64, 128, 256, 512) → Global Average Pooling	Dense(1, sigmoid)

setting were employed: Vanilla Gradients (GRAD), Integrated Gradients (IG), and Smooth Gradients (SG). Each of these methods produces a relevance matrix that assigns importance scores to specific characteristics at each time point, thereby identifying which segments of the input have the greatest impact on the model's decision. For instance, saliency analysis might reveal that increased quiz activity during the third week of a course is a decisive predictor of student success. This temporal resolution is especially critical in educational contexts, where discerning not only which behavioural attributes are influential but also precisely when they exert the most influence enables the formulation of timely and targeted interventions.

The reliability of temporal attribution methods was assessed using the perturbation-based validation protocol listed bellow [19]. The full evaluation setup is summarized in Table 4.

1. First, a model is fitted and evaluated quality metrics such as Accuracy, F1 score, and AUC. This initial check ensures that the model works correctly before creating explanations.
2. A time series relevance method generates an explanation for each sample in the test dataset, outlining how every feature and moment influences the predictions.
3. Finally, to test the consistency and reliability of these explanations, perturbation and changes are applied to the original data according to their explanations. After this, the performance of the model is subjected to an additional evaluation, and the conditions $qm(t) \geq qm(tcr) > qm(tc)$ must be satisfied to confirm accurate explanations from the XAI method. Where t represents the original time series, tcr represents a randomly modified time series, and tc represents the time series modified according to the time series relevance method. This condition ensures that the explanations truly reflect the significance of the input features in the model prediction.

This study uses the following perturbation methods:

- Zero: given a time series $t = (t_0, t_1, t_2, ..., t_m)$ and the relevance of its features $r = (r_0, r_1, r_2, ..., r_m)$, the features with relevance above a threshold ϵ are replaced with zeros to observe how the removal of critical characteristics affects the performance of the model.

- Inverse: similar to the previous perturbation method, it identifies the critical features that exceed a threshold and replaces them with their inverse values. For example, if $t_{sub} = (t_i, t_{i+1}, ..., t_{i+ns})$ is the extracted subsequence based on relevance $r_i > \epsilon$, it is replaced by $t_{sub}^{-1} = (t_i^{-1}, t_{i+1}^{-1}, ..., t_{i+ns}^{-1})$ and then this new subsequence is inserted back into the time series.

- Swap: this perturbation extracts the subsequence of relevant features and replaces each with additional random features from the input.

- Noise: in this perturbation, the relevant subsequence $t_{sub} = (t_i, t_{i+1}, ..., t_{i+ns})$ is modified by adding random noise, $t_{sub} + \eta$ where η is a random variable.

- Mean: in this method, the mean $\mu_{t_{sub}}$ of the subsequence is calculated and the entire subsequence is replaced by this mean value, $t_{sub} = (\mu_{t_{sub}}, \mu_{t_{sub}}, ..., \mu_{t_{sub}})$.

The baseline is established using random features and moments. This procedure is then applied to validate that the time series relevance method provides robust explanations and accurately captures the importance of the time series.

Table 4: Perturbation-based validation protocol for temporal attribution methods.

Item	Value
Attribution methods	IG, GRAD, SG, Random baseline
Implementation	TSInterpret [42]
Perturbation operators	zero, inverse, noise, swap, mean
Perturbed region	Top-p percentile of relevance scores (per sample)
Percentile p	0.9
Noise parameters	Gaussian noise with $\sigma = 0.5$
Evaluation splits	5 randomized train/test splits
Reliability criterion	Performance drop

5.4. Counterfactual Baselines

Two state-of-the-art time-series counterfactual methods are used as baselines. CoMTE is a model-agnostic optimization-based approach that minimizes the distance between original and counterfactual sequences under temporal constraints. TSEvo is an evolutionary algorithm that balances prediction validity, sparsity, and plausibility.

Neither baseline incorporates attribution information to guide where perturbations occur. This allows a direct evaluation of the added value of saliency-guided counterfactual generation.

5.4.1. Evaluation Criteria

Following the evaluation principles in [10], each counterfactual is evaluated using four dimensions:

- Sparsity: the number of changes from the original instance. It is a crucial metric because counterfactuals should be as simple as possible, unless an apparent reason dictates otherwise, altering only the necessary features to achieve the desired change in prediction. Ensuring simplicity helps teachers and administrators understand and effectively implement personalized recommendations for each student.

- Similarity: the distance between the generated counterfactual and the original sample remains close. This proximity makes the counterfactuals generated reasonable and appropriate for use in real-world settings. The Dynamic Time Warping (DTW) distance [43], which is specifically designed for time series, is used to measure similarity.

- Plausibility: the distance between the generated counterfactual and the other samples. This metric ensures that counterfactuals align with realistic and coherent scenarios in the educational context. To ensure the effectiveness and consistency of the counterfactuals generated based on the behaviour of other successful students, the final distance is measured as the average distance between the five closest real students.

- Probability Shift: the increase in predicted probability toward the desired class after applying the counterfactual changes. In educational settings, this reflects the significant improvement in the probability of passing. A higher shift indicates a more convincing and robust counterfactual, as it suggests greater confidence in the student's progress after the proposed changes.

These metrics ensure both technical strength and pedagogical relevance,

providing explanations that are feasible, interpretable, and meaningful in real-world settings.

5.4.2. Generalizability

Although evaluation is conducted on OULAD, the proposed framework is not dataset-specific. The methodology can be applied to other forms of educational time-series data, such as logs from learning management systems or massive open online courses, provided suitable predictive models and attribution techniques are available.

5.5. Implementation Details

All experiments were conducted under a fixed hardware configuration to ensure comparability across models and explanation methods. Table 5 summarizes the experimental environment.

Table 5: Experimental environment.

Component	Specification
CPU	Intel Core i7-9700 (8 cores, up to 4.7 GHz)
RAM	62 GB
GPU	NVIDIA GeForce RTX 2080 Ti
GPU driver / CUDA	470.182.03 / CUDA 11.4
Operating system	Linux (x86_64)

6. Results

This section reports the empirical evaluation of the proposed framework. The study encompasses four steps: 1) the comparison of predictive perfor-

mance of used deep learning models for early risk prediction, 2) the quantitative comparison of gradient-based temporal attribution methods to justify the usage of a saliency technique for steering counterfactual generation, 3) the quantitative comparison of counterfactual generation methods, and 4) the qualitative feedback from instructors in an exploratory way to perceive the interpretability and usefulness.

6.1. Predictive Performance of Deep Learning Models

Table 6 reports mean performance across five stratified cross-validation folds. However, the comparison is intended to inform the selection of a representative predictive backbone for subsequent explainability methods rather than to establish definitive performance rankings.

Table 6: Results of deep learning prediction models

Model	**Accuracy**	**AUC**	**F1-score**
ResNet	0.769	0.85	0.766
CNN	0.758	0.845	0.771
LSTM	0.767	0.851	0.759
BiLSTM	**0.771**	**0.853**	**0.772**
ConvLSTM	0.743	0.834	0.761

Among the models tested, the BiLSTM model yielded the highest scores (Accuracy = 0.771; F1 = 0.772; AUC = 0.853), suggesting that this architecture is particularly effective in capturing temporal dependencies in both directions. This bidirectional capability likely enhances the detection of nuanced patterns in student engagement over time, which is crucial for early

identification of risk trajectories. Nevertheless, other model architectures, such as CNNs, LSTMs, and ResNet, also produced promising scores, confirming the presence of sufficient predictive signal in student action data. These findings underscore that multiple deep learning approaches can leverage temporal and behavioral information in the data, which is essential for developing early-warning systems that can support timely educational interventions.

6.2. Reliability of Temporal Attribution Methods

To evaluate which gradient-based attribution method most reliably highlights influential input features, the reliability of the attribution methods was assessed using a perturbation-based procedure. In this process, the relevant features identified by the different approaches were perturbed, and each perturbation was followed by a measurement of the resulting degradation in predictive performance.

The decreases in Accuracy, F1-score, and AUC with respect to the five perturbation strategies (inverse, mean, noise, swap, and zero) are listed in Tables 7, 8, and 9. A larger decrease indicates that the attribution method successfully pinpoints features and time points relevant to the model's predictions.

Table 7: Accuracy Decrease

XAI Method	inverse	mean	noise	swap	zero
GRAD	**0.1626**	0.018	**0.2541**	0.0248	0.0144
IG	0.1609	**0.0206**	0.251	**0.034**	**0.0167**
SG	0.0000	0.0000	0.0000	0.0000	0.0000
Random	0.1349	0.0051	0.2227	0.007	-0.0013

First, Table 7 presents the decreases in accuracy for all methods. These results indicate that both IG and GRAD find relevant features for the model. The most significant reductions for IG are 0.0206 in "mean" perturbation, 0.034 in "swap", and 0.0167 in "zero", which indicates that IG is especially effective in identifying critical moments that affect model accuracy. GRAD also presents notable decreases, with values of 0.1626 in "inverse" and 0.2541 in "noise". On the other hand, SG does not show any reduction in accuracy within different perturbations. The reason could be that this method significantly smooths the gradients, and critical perturbations are not reflected accordingly.

Table 8: F1 Decrease

XAI Method	inverse	mean	noise	swap	zero
GRAD	**0.3394**	0.0329	**0.5019**	0.0440	**0.014**
IG	0.3367	**0.0382**	0.4976	**0.0583**	**0.014**
SG	0.0000	0.0000	0.0000	0.0000	0.0000
Random	0.2905	0.0098	0.4485	0.0124	-0.0057

Meanwhile, the results of the reductions in F1 are listed in Table 8. In

this metric, the results align with the findings in accuracy, underscoring the strength of IG and GRAD to identify the correct features and timestamps. IG produces a reduction of 0.0382 in mean" and 0.0583 in swap'', while the decreases produced by GRAD are 0.3394 in inverse" and 0.5019 in noise". SG still does not produce any reduction and confirms that, at least for OULAD dataset, its smoothing process has a detrimental effect on its ability to capture different importances in the time series.

Table 9: AUC Decrease

XAI Method	**inverse**	**mean**	**noise**	**swap**	**zero**
GRAD	**0.2248**	0.0936	**0.3158**	0.0997	0.1001
IG	0.2232	**0.0957**	0.3127	**0.1086**	**0.1028**
SG	0.0783	0.0783	0.0783	0.0783	0.0783
Random	0.1976	0.0824	0.2841	0.0842	0.0779

Finally, Table 9 presents the results of the AUC reduction for the techniques evaluated. These results finally validate the usefulness of IG and GRAD in identifying critical moments. However, SG cannot improve upon the baseline, indicating that either excessive smoothing impairs SG's ability to recognize crucial events or fails to effectively capture perturbation effects.

Additionally, a post-hoc Nemenyi test (Figure 3) was performed. This test showed that differences between methods were significant, with IG having the lowest average rank over all metrics. Therefore, the statistically better-performing attribution method was Integrated Gradients, chosen for its guidance on counterfactual generation.

Finally, a qualitative showcase for a well-performing student is presented,



Figure 3: Nemenyi test for Accuracy.

referred to as "Good Performance" and for an under-performing student, referred to as "At-Risk". The purpose of this showcase is to illustrate how each method highlights the most significant moments in the student's activities, as predicted by the model.

As observed in Figures 4a and 4b, GRAD method indicates that, for a good-performing student (left), the first days of the course appear to be the most relevant. Despite the student's limited interaction on many of these days, GRAD suggests that an early, strong start is crucial to the student's success. On the other hand, the results obtained for the at-risk student demonstrate a progressive disconnection, as evidenced by the peaks being concentrated on days when his activity decreases, particularly on days 28 to 35. This finding underscores the importance of progressive disconnection in his negative prediction. However, GRAD demonstrates sensitivity to minor variations in the data, reflecting instantaneous reactivity. For instance, pronounced colour changes and narrow bands, such as the variable "Outcontent" for the student who performed well on day 14, suggest that specific events significantly impact the predictions. This capacity makes GRAD a valuable tool for identifying critical moments; however, its reactivity makes it challenging to generalize its findings to irregular learning trajectories.

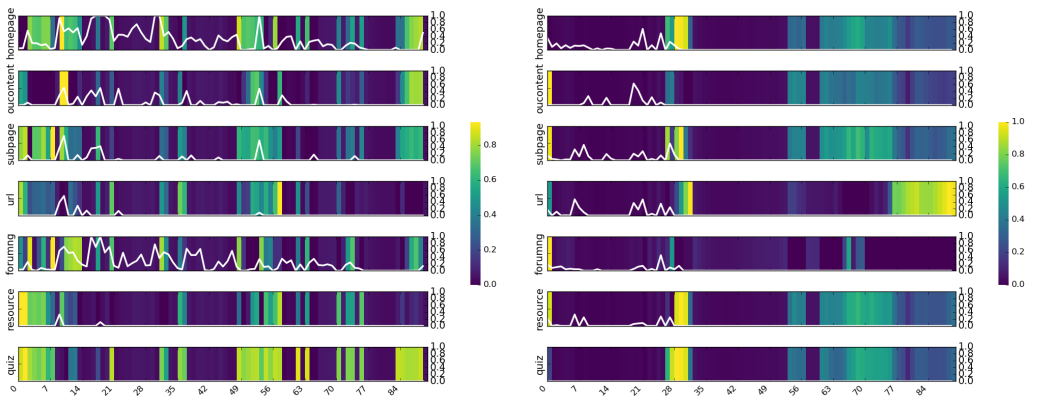
Figures 4c and 4d provide the explanations obtained with IG. Initially, these explanations are highly similar to GRAD explanations. However, a

closer examination reveals that the colour bands appear less pronounced and gradually increase in importance over time, suggesting a less reactive influence on feature changes. For the student who performs well, IG highlights the importance of the last days, particularly in features such as "HomePage", "Resource" or "Quiz". This emphasis suggests that continuous participation throughout the course is crucial to ensuring academic success. The most informative characteristics of the at-risk student are distributed over time, with notable concentration between days 40 and 56. In this interval, the relevance of the characteristics is lower. Nevertheless, they are more relevant than in GRAD because the method considers this time without interaction important. In summary, IG attenuates oscillations and detects persistent patterns, providing a clearer picture of the cumulative effect of student activity on performance.

Then, Figures 4e and 4f illustrate how SG differs significantly from previous methods. The only relevant feature for both at-risk and good-performing students is "HomePage". It is essential to note that this feature consistently presents values across all days, a consequence of the smoothing process implemented by SG to mitigate noise, resulting in a more constrained approach. Unlike GRAD and IG, which reveal detailed patterns, SG is more selective in identifying critical variables, eliminating much of the variability. However, this smoothing process can be excessive, resulting in oversimplification and the omission of key patterns that reflect the complexity of student behaviour.

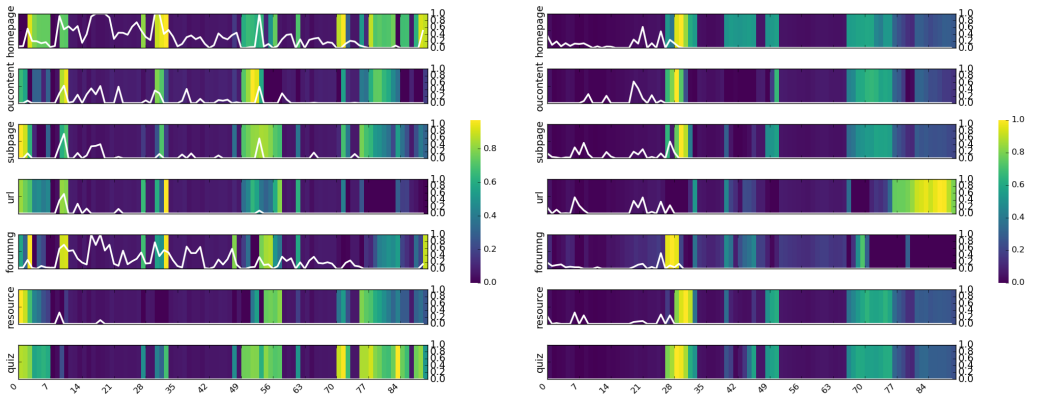
6.3. Quantitative Comparison of Counterfactual Generation Approaches

Once Integrated Gradients was determined to be the best among other attribution methods, its relevance scores were applied to generate counterfac-



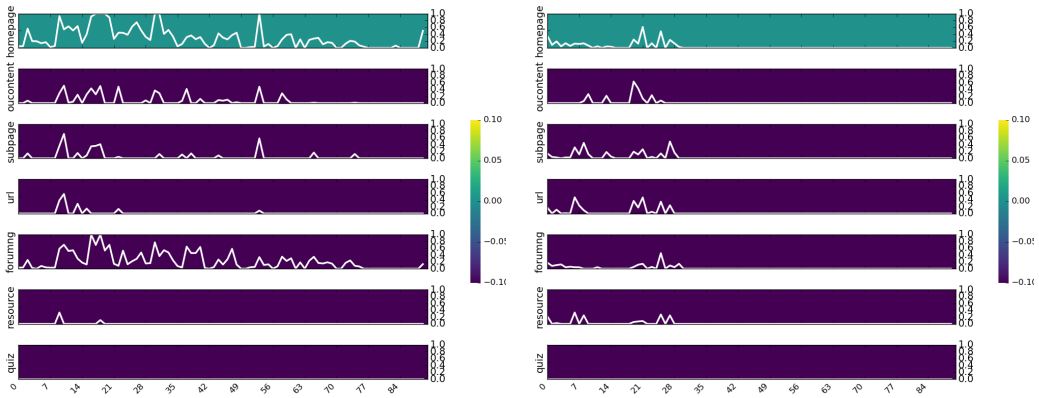
(a) Good Performance (Vanilla Gradients)

(b) At-Risk (Vanilla Gradients)



(c) Good Performance (Integrated Gradients)

(d) At-Risk (Integrated Gradients)



(e) Good Performance (Smooth Gradients)

(f) At-Risk (Smooth Gradients)

Figure 4: Examples of explanations generated by different attribution methods.

tuals for a sample of vulnerable students. The following counterfactual approaches were compared: CoMTE, the regular TSEvo, and the two saliency-based methods proposed in this paper: Guided Local Search Counterfactuals (LSCF) and Guided TSEvo. The guided methods used thresholds based on percentiles (95th, 97th, and 99th) of saliency scores to limit the number of changes that could be made. In practical applications, percentile selection reflects a trade-off between interpretability and coverage. Higher percentiles tend to produce more localized and cognitively manageable explanations suitable for case-based instructor reflection, whereas lower percentiles may be preferred for exploratory analyses that prioritize broader behavioural coverage.

Table 10 reports the results of different counterfactual generation strategies, assessed through five criteria: sparsity, similarity, plausibility, computational time, and probability shift. However, given the high computational cost of TSEvo, the number of training epochs was reduced from 500 to 100 to improve efficiency. To ensure fairness, all methods were evaluated using the same pool of 2000 at-risk students.

Table 10: Counterfactuals Comparison

Method	Sparsity	Similarity	Plausibility
COMTE	20.42 ± 14.73	0.38 ± 0.22	**0.83 ± 0.39**
LSCF 95	28.72 ± 6.48	0.58 ± 0.11	1.20 ± 0.41
LSCF 97	18.2 ± 2.28	0.43 ± 0.06	1.07 ± 0.40
LSCF 99	6.95 ± 0.23	**0.22 ± 0.08**	0.94 ± 0.41
TSEvo	**6.54 ± 4.42**	0.23 ± 0.15	0.91 ± 0.41
TSEvo-Guided 95	7.71 ± 5.43	0.24 ± 0.11	0.89 ± 0.40
TSEvo-Guided 97	7.83 ± 4.85	0.25 ± 0.11	0.90 ± 0.39
TSEvo-Guided 99	7.12 ± 4.23	0.24 ± 0.11	0.89 ± 0.40

Method	Time	Probability Shift
COMTE	40.75 ± 0.75	0.16 ± 0.09
LSCF 95	42.47 ± 1.44	**0.18 ± 0.11**
LSCF 97	41.73 ± 5.59	0.15 ± 0.11
LSCF 99	**39.63 ± 0.79**	0.10 ± 0.09
TSEvo	1421.27 ± 23.97	0.0757 ± 0.07
TSEvo-Guided 95	1387.79 ± 73.52	0.07 ± 0.08
TSEvo-Guided 97	1383.57 ± 75.81	0.07 ± 0.08
TSEvo-Guided 99	1481.36 ± 53.56	0.07 ± 0.07

In terms of sparsity, TSEvo and its guided variants required the fewest modifications (6.5–7.8), with LSCF 99 very close (6.96). The differences are

not statistically significant (Figure 5a), but the result confirms that LSCF can suggest interventions with minimal disruption of the trajectory of a learner.

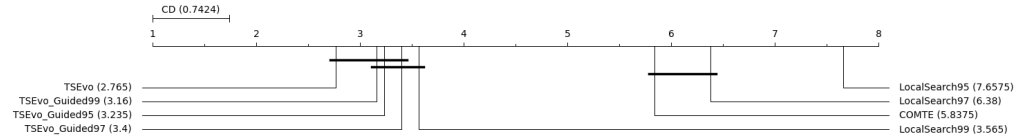
Similarity tells a different story (Figure 5b). Here, LSCF 99 produced the closest counterfactuals to the original sequences (0.22), which is slightly better than TSEvo (0.24–0.25) and significantly ahead of COMTE (0.38). This margin, though small, suggests that LSCF captures subtle changes that feel natural rather than artificial, an essential quality in education where unrealistic shifts may confuse rather than guide.

Plausibility favours COMTE, which achieved a score of 0.83. This indicates that its counterfactuals resemble real student patterns more closely than those produced by LSCF. The difference is statistically significant (Figure 5c). Still, plausibility should not always dominate. For early warning tools, interventions that are sparse and interpretable may be more useful than those that mirror historical trajectories.

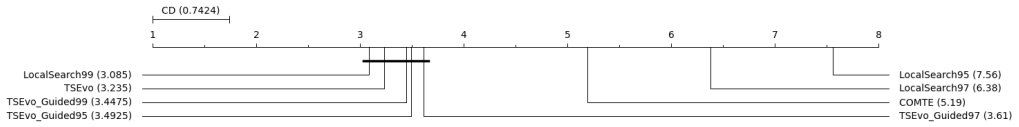
The probability shift highlights the strength the local search. LSCF 95 and 97 outperformed every other approach, including COMTE and TSEvo (Figure 5d), demonstrating that even small, well-targeted changes can significantly improve prediction to success.

Computational efficiency showed the strongest differentiation between the approaches. Local Search and CoMTE methods required around 40s per instance to produce counterfactuals, while methods based on TSEvo demanded more than 1,400s. This discrepancy makes evolutionary methodologies unfit for online applications, while local search methods can be more easily adapted to time-pressured learning contexts. In summary, LSCF 99 provides a reasonable compromise across sparsity, similarity, probability shift, and

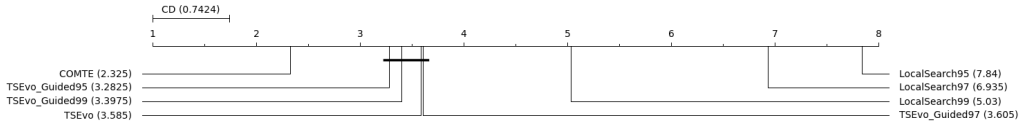
computational complexity, making it appropriate for interpretability, counterfactual explanations, and temporal coherence.



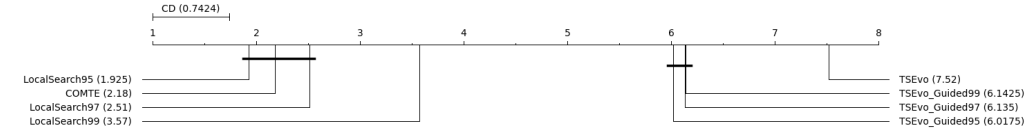
(a) Sparsity



(b) Similarity



(c) Plausibility



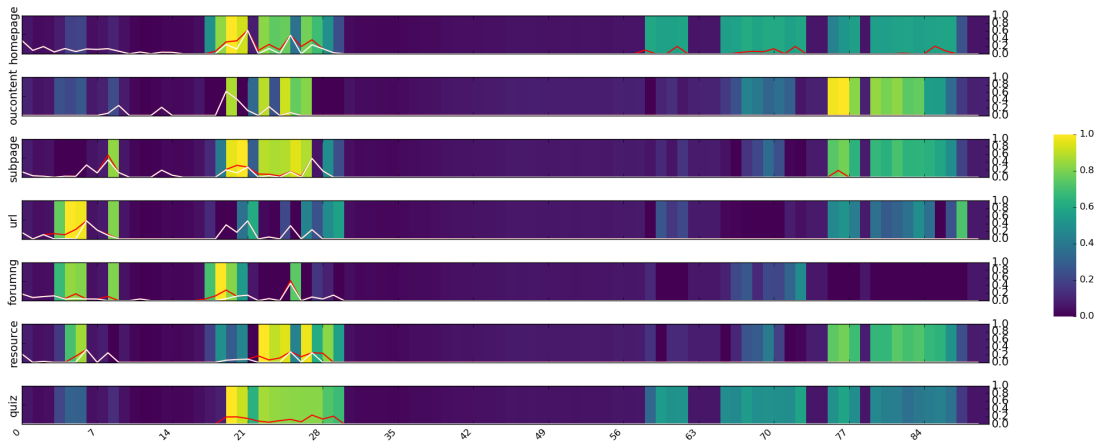
(d) Probability Shift

Figure 5: Nemenyi test results across key counterfactual evaluation metrics. A lower average rank indicates better performance.

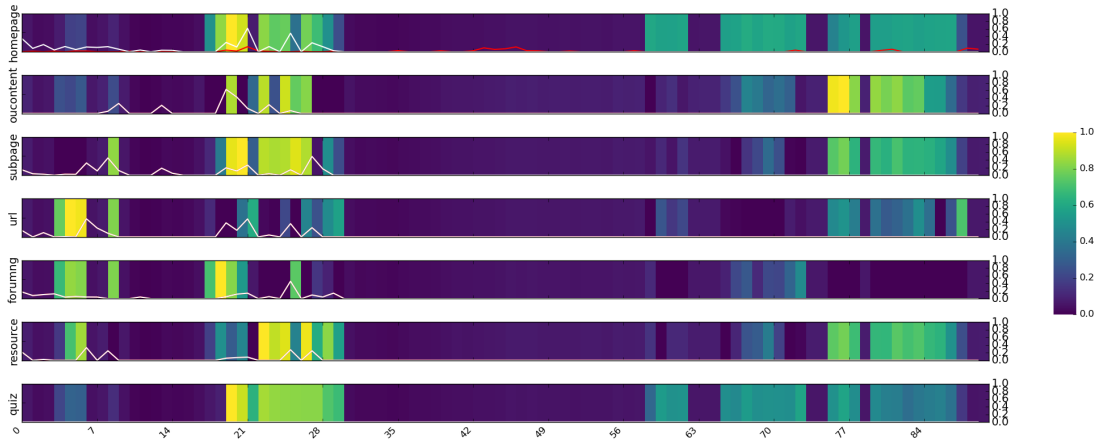
Finally, Figure 6a illustrates a counterfactual example generated by LSCF for the "At-Risk" student shown earlier in Figures 4b, 4d, and 4f. In the plot, red lines indicate the modified interaction patterns suggested by LSCF, while white lines show the original trajectory. The contrast highlights how small, well-placed changes can shift the model prediction away from risk.

The counterfactual primarily adjusts variables such as *HomePage* and *Quiz*, particularly within the critical time windows identified by IG. From an instructional perspective, this counterfactual highlights potential moments and behaviours for pedagogical reflection that more engage with the course after day 60 and to complete the quizzes skipped between days 20 and 30 could be sufficient to modify the model's risk assessment, providing a concrete illustration of how the predictive system associates specific temporal behaviors with outcome probabilities. Nonetheless, the suggested changes should be interpreted as hypotheses that require contextual validation by instructors.

In comparison, the counterfactual created with COMTE (Figure 6b) is less informative. It only suggests increasing homepage visits and makes changes at times when the attribution analysis deems them unimportant. More importantly, it does not meet the needs of real classrooms, since just revisiting the homepage without engaging with assessments or content is unlikely to help students improve. In contrast, LSCF provides more informative and pedagogically meaningful signals to support educator reflection.



(a) Counterfactual generated by LSCF



(b) Counterfactual generated by COMTE.

Figure 6: Comparison of counterfactuals for the same student instance. The original sequence is shown in white, and the counterfactual modifications in red.

6.4. Instructor Qualitative Exploratory Feedback

Quantitative evaluation metrics alone are insufficient to capture how instructors perceive and interpret explanation methods in real educational contexts. For this reason, an exploratory qualitative study was conducted re-

garding the interpretability and perceived potential usefulness of the proposed temporally grounded counterfactual explanations. The purpose of this study was not to evaluate instructional effectiveness, behavioural change, or learning outcomes, but rather to examine whether the explanations are understandable, appear plausible, and can support instructor reflection on student engagement trajectories.

The sample consisted of 20 university teachers from the University of Cordoba, and the instrument comprised a structured Likert-scale questionnaire and an unstructured questionnaire. The respondents were shown visual saliency maps, counterfactual trajectories, and text explanations generated through the framework. The demographics of the respondents are shown in Table 11.

The questionnaire attempted to cover four dimensions of the explainability pipeline, namely: (i) comprehension of the temporal impact on model outcomes; (ii) users' assessed utility of visual mode of rendering explanations; (iii) understandability and authenticity of counterfactual hypotheses; (iv) users' comparative perceptions of textual explanations, contrasting instructor-authored descriptions with those generated by a multimodal language model (LLaVA) [44]. There were two prompting approaches to be explored for the language model: 'zero-shot' design, and other based on oriented prompting.

Table 11: Demographic Profile of Survey Participants

Characteristic	**Category**	**Count**
Role	Assistant Professor	10
	Associate Professor	6
	Full Professor	4
Gender	Female	5
	Male	15
Age	25-35 years	6
	35+ years	14
Familiarity with AI	None	3
	Basic	10
	Intermediate or higher	7

Example materials and the full survey instrument are freely accessible online at: `https://forms.gle/f3kQwtDoq3aSRSsn9`. This aligns with the increasing push in the XAI literature for user-centred evaluations in practical domains such as education [45].

Across models, participants reported that explanations emphasizing the temporal aspects of student behavior facilitated their understanding of model predictions. Many referenced that temporal visualizations were beneficial for pinpointing when patterns of student disengagement began or escalated. Features associated with assessment activities, content viewing, and engagement with the course homepage were consistently cited as intuitive measures of student engagement (Figure 7). Some instructors indicated that temporal

saliency maps were consistent with their experiences, while other instructors mentioned that the explanations uncovered patterns they would have overlooked using only aggregate indicators.

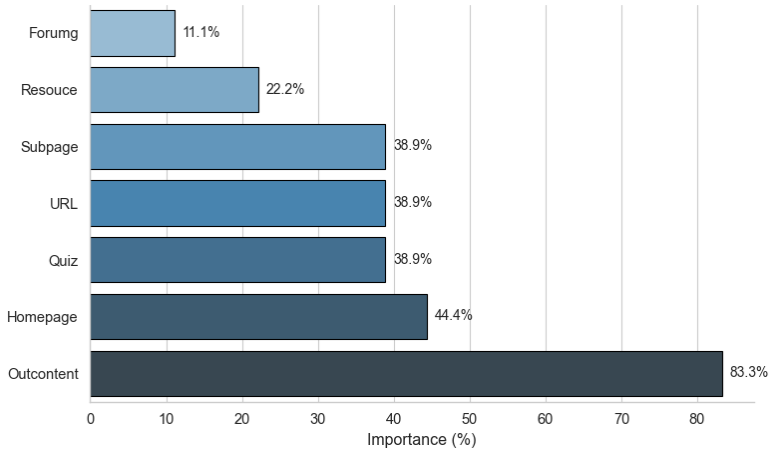


Figure 7: Expert-perceived importance of input features.

Overall, explanations based on visualizations of outcomes were considered more helpful for interpretation than static summaries, particularly when the relevant information was concentrated on specific time periods. Aggregated usefulness ratings for the considered explanation types are reported in Figure 8. Counterfactual trajectories limited to salient time intervals were frequently cited as easier to understand than those with overly general or diffuse changes. Among the considered methods, the guided local search and the guided evolutionary approaches were viewed as providing more concise trajectories, which participants associated with fewer overgeneralized or unrealistic alterations (Figure 9). Meanwhile, participants did not hesitate to stress that they should never be treated as firm recommendations, but only as complements to professional judgment.

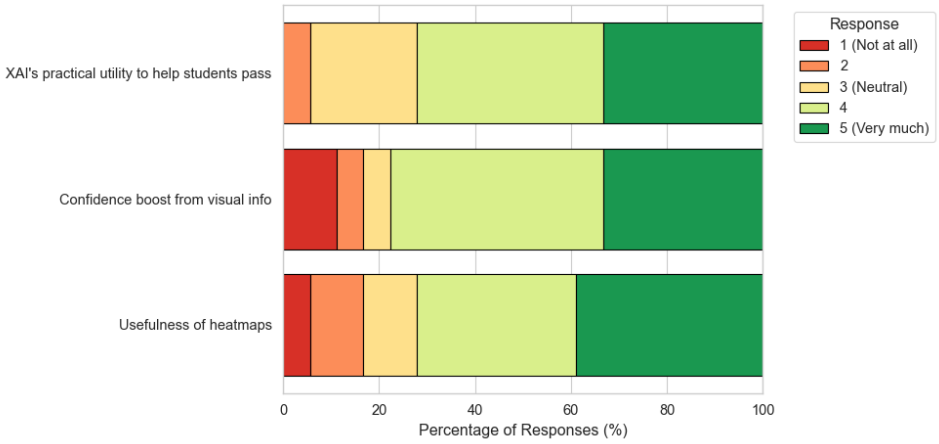


Figure 8: User ratings of usefulness for XAI visual explanations.

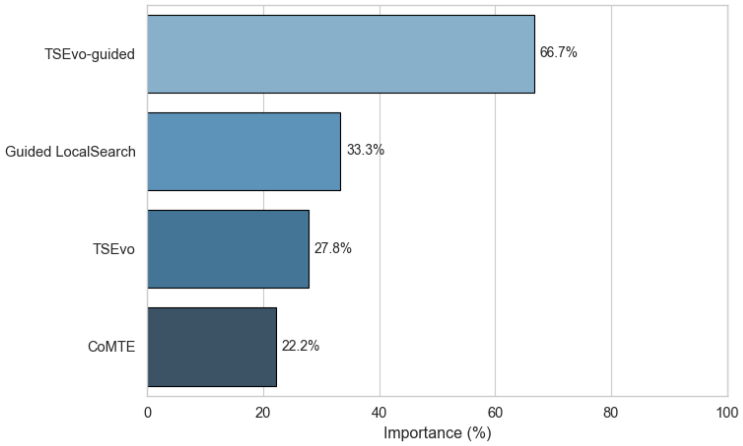


Figure 9: Preference ranking of counterfactual methods by participants.

In general, instructor-provided textual explanations were rated as clearer and more context-sensitive than automatically generated ones. However, the learners viewed descriptions from LLaVA under pedagogy-prompted conditions as a potentially valuable supplementary resource in environments where scaling personalized feedback poses challenges. Zero-shot generations were

largely characterized as generic and lacking in context, consistent with previous research on the application of language models to human-centric explanation work [46]. (Figure 10).

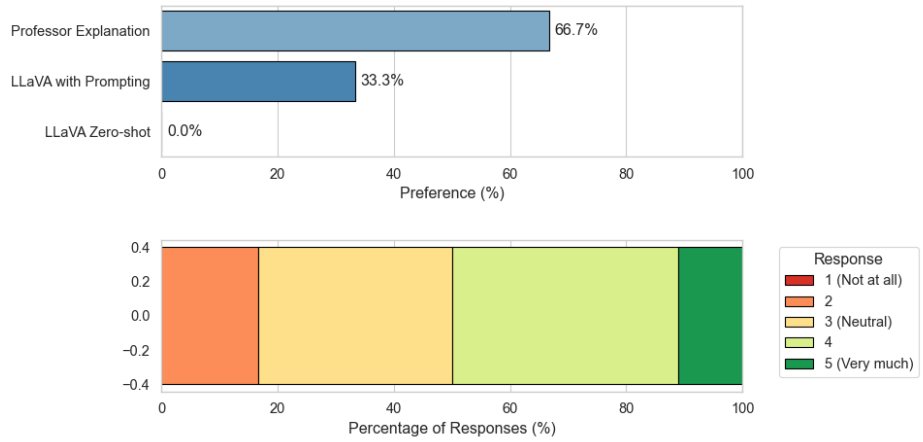


Figure 10: Comparison of LLM and human-generated explanation ratings.

The qualitative feedback indicates that instructors found temporally grounded, visual, and counterfactual explanations interpretable and aligned with their reasoning about student engagement when these explanations were framed as analytic supports rather than prescriptive recommendations. However, due to the limited scope and sample size of the present study, this evidence should be taken as suggestive rather than definitive and warrants follow-up, large-scale, diversified, user-centered assessments.

7. Discussion

This work suggests that combining temporal attribution techniques with counterfactual generation can enhance the interpretability of educational

time-series predictive models. Early warning systems based on deep learning achieve strong predictive performance, but to be used in the real world, teachers require timely, clear, and prospective explanations. The method described in this work seeks to address these issues by evaluating the model's sensitivity to timing and by guiding the analysis of which student behaviors affect the prediction.

Testing the influence of individual features on predictions reveals that Integrated Gradients provides a consistent, interpretable signal from the BiLSTM. Vanilla Gradients also identify significant events and behaviors, but their susceptibility to perturbation precludes their use in the counterfactual method. Smooth Gradients perform poorly, likely because over-smoothing in student data obscures salient events and behaviors. These findings reinforce existing research that identifies Integrated Gradients as most suitable for balancing trustworthiness and interpretability, especially when the objective of explanations is to support educators' reasoning rather than merely inspecting predictions [47] [48] [49].

The comparison between guided counterfactuals and descriptive results indicates that the former provides concise counterfactual scenarios that highlight critical time periods and are more closely aligned with observed student interaction patterns. Moreover, tests with additional saliency percentile thresholds suggest that the choice of threshold allows calibration of interpretability and predictive effectiveness. Lower thresholds allow wider changes and, generally, greater probability gain; higher thresholds yield shorter, more interpretable, but less effective, explanations. These thresholds can be interpreted as design choices that trade off interpretability and breadth of

change. From an analytical perspective, when researchers need to quickly discover broad behavioral patterns, lower thresholds are preferable. In contrast, higher thresholds should be used when the specific case demands greater interpretability, such as in case studies or post-hoc analyses focused on variable effects. The explanations are intentionally concise, prioritizing rapid interpretability over exhaustive reconstruction of student behaviour.

Finally, computational complexity is relevant for large-scale deployment in real-world scenarios. The counterfactual-based evolutionary methods provide highly sparse explanations. However, their high computational complexity precludes their use in environments requiring prompt responses. The local search strategy, on the other hand, produces competitive-quality explanations much more quickly, enabling its use in workflows that require prompt explanations. This result illustrates the need to assess explainability techniques considering not only technical but also operational aspects in educational environments.

Exploratory qualitative feedback from instructors indicated that temporally localized explanations were often perceived as more interpretable than global or static representations. The ability of visualizations to access time dimensions was favored. They were able to demonstrate the time at which patterns of disengagement emerged, as well as the time at which they could exert their possible effects. Counterfactual explanations that focused on a limited number of impactful dimensions were also perceived as easier to interpret by instructors. Recommendations from other interviews carried a disclaimer. Instructors argued that these types of explanations should complement, rather than dictate, their professional judgment. This observation

aligns with the notion that explainable learning analytics systems should function as decision-support, not as decision-making systems with automated interventions.

In a teaching context, text summaries are perceived as useful, though they offer some benefit; nevertheless, human-produced explanations are rated as more comprehensible and student-centered. Overall, these tools may assist educators in outlining explanations at scale, while human-in-the-loop oversight remains essential to add professional judgment to the results. Hence, human-centered investigations are also essential to determine the impact of such explanations on trust and practical use.

To overcome this problem, the framework should be tested on datasets from different institutions and platforms in future studies. Such an approach may provide a better insight into its generalizability and utility for other settings.

The qualitative aspect of the current study was also limited. It was conducted with a rather small and homogenous group of instructors, which limits the utility of qualitative findings. Therefore, future studies may involve larger, more diverse samples, including instructors with different expertise, positions, or geographical locations, to provide more generalizable conclusions.

Finally, the current evaluation was focused on counterfactual validity, sparsity, and interpretability of the explanation, without exploring whether these attributes affect instructional choices and student outcomes. Addressing this limitation in future studies may be achieved through conducting longitudinal interventionist projects that integrate explainable analytics into

authentic educational contexts and investigate their educational value and effects.

In this context, it is also important to acknowledge the risk that counterfactual explanations may highlight behavioural changes that improve model predictions without necessarily addressing deeper learning processes. Such explanations should not be interpreted as prescriptions for optimizing student behaviour, but rather as hypotheses to support instructor reflection and sensemaking. This limitation reinforces the need for human judgment and pedagogical expertise when translating model-based explanations into educational action.

To sum up, time series in education is irregular and strongly contextualized. Many factors affect them, including cognitive and motivational mechanisms and learning situations that are not observable in platform logs. This circumstance limits to some extent the reach of any model-like explanation in capturing the learning process. The framework proposed here does not address this situation. It seeks to provide a framework for connecting complex predictive models with human reasoning in an orderly manner. This contribution brings together temporal attribution and counterfactual reasoning as a framework for supporting human interpretation of complex predictive models in educational contexts. It provides one more step towards explainable learning analytics systems, which account for empirical results and acknowledge the limits of the processes underlying educational decision-making.

8. Conclusions and Future Work

This study presented two saliency-guided methodologies for generating counterfactual explanations in multivariate educational time series. The proposed framework integrates gradient-based temporal relevance analysis with counterfactual generation to support the interpretability of early risk predictions derived from student interaction data.

The quantitative analyses suggest that Integrated Gradients provides a reliable and informative attribution signal for modelling student learning trajectories and is more robust to perturbations than other gradient-based approaches. While Vanilla Gradients may still be useful for exploratory analysis, their sensitivity to small variations limits their suitability as a stable guiding signal for counterfactual generation in educational settings. Smooth Gradients appeared less appropriate in this context, potentially due to over-smoothing effects that obscure sparse but pedagogically relevant temporal events.

Two saliency-guided counterfactual generation strategies were introduced and evaluated. The local search–based approach (LSCF) generated compact and temporally coherent counterfactual trajectories, balancing sparsity, proximity to the original data, probability change, and computational efficiency. The guided evolutionary approach (Guided-TSEvo) produced similarly sparse alternatives but at substantially higher computational cost, which may limit its applicability in time-sensitive or large-scale educational workflows. These findings highlight the importance of considering operational constraints, such as explanation latency, alongside counterfactual quality when designing explainable learning analytics systems.

Exploratory feedback from instructors indicated that counterfactual explanations grounded in specific time periods were perceived as easier to interpret than more diffuse or non-specific alternatives. Temporally localized counterfactual paths were described as helpful for reflecting on when changes in student engagement may have influenced model predictions. However, as the qualitative component of this study was limited in scope and did not assess instructional impact or learning outcomes, these observations should be interpreted with caution.

Several limitations should be acknowledged. The framework was evaluated on a single dataset from one institutional context, and the qualitative study involved a relatively small and homogeneous group of instructors. Future work should examine the generalizability of the proposed approach across different educational platforms, disciplines, learner populations, and instructional settings. Additionally, further research is needed to investigate how counterfactual explanations are integrated into authentic teaching practices and whether they influence instructional reasoning or decision-making over time.

Future research directions include incorporating richer pedagogical and contextual constraints into counterfactual generation, exploring alternative temporal representations of student behaviour, and further investigating the role of language-based models as mediating technologies that translate technical explanations into educator-oriented narratives. Advancements along these lines may contribute to more human-centred explainable learning analytics systems in which predictive modelling is better aligned with instructor reasoning, reflection, and professional judgment.

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Declaration of generative AI

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Author Contributions

All authors contributed to the development of the study and the writing of the manuscript. All authors have read and approved the final version of the paper.

Highlights

- Integrated Gradients outperforms other methods for time-series attribution
- Two new counterfactual algorithms: LSCF and TSEvo-Guided are proposed
- LSCF yields sparse, realistic, and fast counterfactual recommendations

- TSEvo-Guided offers clear suggestions but is limited by high computation time

- Educators valued IG and guided counterfactuals for targeted student support

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: